

MATH503 week 7 tutorial

Instructions:

Find the solutions for the following questions

- First using a Graphics Calculator or online Normal Distribution Calculator and/or Inverse Normal Calculator

Link for graphics calculator emulator:

<https://www.intereseducation.com/resources/casio-fx-9860g-sd-graphing-calculator-emulator-full-version/>

Link for Normal Distribution Calculator:

https://onlinestatbook.com/2/calculators/normal_dist.html

Link for Inverse Normal Calculator:

https://onlinestatbook.com/2/calculators/inverse_normal_dist.html

- Then using RStudio to edit r

pnorm(x,mean,sd) gives the area to the left of x

qnorm(area, mean, sd) gives the boundary value when given the area to the left of the boundary value

rnorm(n, mean, sd) gives n random samples from the normal distribution

Tutorial questions:

1. The heights of a population of young women are normally distributed with a mean of 167 cm and standard deviation 5 cm. Calculate the probability that a student chosen at random has a height
 - a) between 161cm and 173cm. $P(161 < X < 173)$
 - b) between 155cm and 185cm. $P(155 < X < 185)$
 - c) less than 167cm. $P(X < 167)$

d) greater than 162cm. $P(X > 162)$

e) between 149cm and 173cm.

f) greater than 173

2. The water roller coaster has a minimum height requirement of 1.52m. The average height of children who visit this amusement park is 1.22m with a standard deviation of 0.61m. Assume that the distribution of children's heights is approximately normal.
- a) Approximately what percentage of the children who come to the park qualify to ride the roller coaster.
 - b) If it is known that 850 children visited the park on a given day, how many of these children were able to ride the roller coaster?
3. Mensa is an organization that allows people to join only if their IQs are in the top 2% of the population. What is the lowest Stanford-Binet IQ you could have and still be eligible to join Mensa if the mean is 100 and standard deviation is 16?
4. An energy-saver light bulb is advertised as having an average life of 10,000 hours with a standard deviation of 700 hours. The factory produces 100,000 bulbs per day. If the life spans are normally distributed:
- a) Below how many hours will the weakest 10% of bulbs last?
 - b) Above how many hours will the strongest 5% of bulbs last?
 - c) Between how many hours will the central 80% of bulbs last?